

What are motives for?

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The first of the “standard conjectures” (Grothendieck [19], Kleiman [20]), the conjecture of Lefschetz type, asks that certain cohomology classes be “direct” images of algebraic cycles. In any case, if motives are the “direct factors” of the cohomology of algebraic varieties X , defined by projectors, namely algebraic cycles on $X \times X$, their definition is reasonable only if one has enough algebraic cycles. On this problem—the construction of interesting algebraic cycles—progress has been meager.

Grothendieck attempted to make a catalogue of projective constructions of cycles (apparent contours), cf. [19, p. 197]. For those whose cohomology class has been computed, this class is expressed in terms of Chern classes of evident vector bundles. Although I know of no counterexample, it seems unlikely to me that the ring of cycle classes on $X \times X$ generated by the diagonal, by divisors, and by the inverse images of the Chern classes of X under the pr_i ($i = 1, 2$), always contains the cycles required by the first standard conjecture, for instance the Kunneth components of the diagonal.

Rather than try to construct cycles, one may try to construct vector bundles and then take their Chern classes. This is in fact how K. Kodaira and D. C. Spencer (1953) prove the Hodge conjecture for divisors (a Lefschetz theorem): the group $\text{Pic}(X) = H^1(X, \mathcal{O}^*)$ is accessible over $\mathbb{C}$ by the exponential exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \rightarrow 0.$$

Similarly, on an abelian variety it is easier to write the functional equation of theta functions (a cocycle for a line bundle) than to define a theta function itself, which would then give the divisor $\theta = 0$.

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Unfortunately, in higher rank we do not know how to construct vector bundles whose Chern classes are interesting any more than we know how to construct interesting cycles.

If $\ell \in \mathbb{C}$, the Hodge conjecture would provide the desired cycles. In the direction of that conjecture, two difficulties have been discovered. The first is that Atiyah and Hirzebruch showed that it can be true only rationally (cf. Atiyah–Hirzebruch [1]).

The second is that, from the viewpoint of Hodge theory, the class of a codimension d cycle Z on a smooth projective X naturally lives not in $H^{d,d} := H^{d,d} \cap H^{2d}$, but in an extension E^d of this group by the intermediate Jacobian

$$J^d(X) := H^{2d-1}(X, \mathbb{C})/F^d + H^{2d-1}(X, \mathbb{Z}),$$

where F^d denotes the d -th term of the Hodge filtration of $H^{2d}(X, \mathbb{C})$.

In the category of mixed Hodge structures, these are respectively $\text{Hom}(\mathbb{Z}(d), H^{2d}(X))$ and $\text{Ext}^1(\mathbb{Z}(d), H^{2d-1}(X))$. The Hodge conjecture asks that the group of cycle classes map to a finite-index subgroup of the quotient $H^{d,d}$. By contrast, we have no idea what the image in E^d should be. In addition to the aesthetic discomfort caused by this ignorance, it generally makes

P. A. Griffiths's method for proving the Hodge conjecture by induction on the dimension of the variety X is inapplicable. This method is inspired by the one Lefschetz used for surfaces, Lefschetz [24]. The idea is that, if $(H_t)_{t \in \mathbf{P}^1}$ is a pencil of hyperplane sections, constructing the cycle Z on X amounts to constructing the cycles $Z_t = Z \cap H_t$. In good cases: (a) the cohomology class $c \in H^{2d}(X)$ that one wants to be the class of a cycle Z has zero restriction to the H_t ; (b) if Z exists, its cohomology class c determines the class of the Z_t in the intermediate Jacobians $J^d(H_t)$; and (c) the construction of Z amounts to the construction of cycles Z_t whose classes define a given section of the family of the $J^d(H_t)$. One can then conclude if every element of $J^d(H_t)$ is the class of a codimension d cycle homologous to zero. By this method, Zucker [32] proves the Hodge conjecture for cubic hypersurfaces in $\mathbf{P}^5$. Note, however, that if every element of $J^d(H_t)$ is the class of a cycle homologous to zero, then $H^{2d-1}(H_t)$ is of Hodge type $\{(d, d-1), (d-1, d)\}$ (see §1.6); the converse is a consequence of the Hodge conjecture applied to the product of H_t with a suitable abelian variety. Moreover, this surjectivity implies the existence of a curve C_t and an algebraic cycle W_t on $H_t \times C_t$ sending $H^1(C_t)$ onto $H^{2d-1}(H_t)$. One can then expect $H^{2d}(X)$ to be controlled by the H^2 of the surface fibered over $\mathbf{P}^1$ with fibers the C_t , and for an H^2 the Hodge conjecture is available in any case.

The purpose of these notes is to show that, despite this absence of progress on the problem of constructing cycles, the philosophy of motives is a powerful tool.

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1 Motives

Depending on what one can and wants to do, one has several definitions of motives—or none. One should distinguish pure motives, typically supplied by the cohomology of nonsingular projective varieties, from mixed motives, where open or singular varieties are allowed. The notion of a motive over S (a family of motives parameterized by S) raises still other problems.

For k a field, in all cases one would like the category of motives over k to be a $\mathbb{Q}$ -linear abelian category $\mathcal{M}(k)$ whose Hom groups are finite-dimensional. In the pure case, one would like it to be semisimple and graded by weight. In the mixed case, every motive M should have a finite increasing filtration W with $\mathrm{Gr}_n^W(M)$ pure of weight n . Other essential structures are:

- (a) Every algebraic variety X over k should have motivic cohomology groups $H_{\mathrm{mot}}^i(X)$, objects of $\mathcal{M}(k)$. In a pure case one restricts to nonsingular projective varieties X , and $H_{\mathrm{mot}}^i(X)$ is a pure motive of weight i .
- (b) For each of the usual cohomology theories H , one should have a “realization” functor real_H and isomorphisms $H^*(X) \xrightarrow{\sim} \mathrm{real}_H H_{\mathrm{mot}}^*(X)$.
- (c) A tensor product $\otimes$, with which the realization functors are compatible.

1.1 Tannakian structure

Structure (c) makes it possible to apply the theory of Tannakian categories, invented by Grothendieck to study the formalism of motives.

References: Saavedra [28], Deligne–Milne [12], Deligne [15].

For k of characteristic 0, this theory says that the category of motives over k , with its tensor product, should be equivalent to the category of linear representations of an affine group scheme G over $\mathbb{Q}$. Note that two groups G_1 and G_2 that are inner forms of one another have the same representation categories. The motivic Galois group G is therefore not uniquely determined by $\mathcal{M}(k)$. The theory says that the choice of G and of an equivalence $\omega : \mathrm{Rep}(G) \xrightarrow{\sim} \mathcal{M}(k)$ is equivalent to the choice of an exact tensor-compatible functor from $\mathcal{M}(k)$ to $\mathbb{Q}$ -vector spaces: a fiber functor ω . To this fiber functor there corresponds $G := \underline{\mathrm{Aut}}^{\otimes}(\omega)$, the group scheme of

tensor automorphisms of ω . One may view ω , or rather (ω, G) , as a “cohomology theory with values in $\mathbb{Q}$ -vector spaces.”

For k equipped with an embedding into $\mathbb{C}$, one possible cohomology theory is: “singular cohomology of the topological space of complex points.” For $k = \mathbb{Q}$, another possible choice is: “de Rham cohomology.”

In characteristic p , an example of Serre shows that there cannot be a reasonable cohomology theory with values in $\mathbb{Q}$ -vector spaces: if E is a supersingular elliptic curve, the algebra $\text{End}(E) \otimes \mathbb{Q}$ is a quaternion algebra, hence has no two-dimensional linear representation over $\mathbb{Q}$. From the existence of a cohomology theory with values in vector spaces over an extension of $\mathbb{Q}$, it follows nevertheless that $\mathcal{M}(k)$ must be the category of representations of a suitable gerbe.

Tannakian theory is a linear analogue of a set-theoretic theory, that of the profinite π_1 , also due to Grothendieck (SGA1). The analogy is this: if $\mathcal{T}$ is the category of finite étale covers of a connected scheme S (respectively, a Tannakian category over K), and ω is a fiber functor with values in finite sets (respectively, in finite-dimensional K -vector spaces), then $\mathcal{T}$ is naturally equivalent to the category of finite G -sets (respectively, of representations of G), for G the profinite group $\text{Aut}(\omega)$ (respectively, the affine group scheme $\underline{\text{Aut}}^\otimes(\omega)$). This is the origin of the terminology “motivic Galois group,” since the theory of π_1 becomes Galois theory in the case of spectra of fields.

1.2 Size of the motivic Galois group

In general, the motivic Galois group G is enormous. Except when k is algebraic over a finite field, one expects, for example, that it has $\text{PGL}(2)$ as a quotient. It does not “know” that motives M may form a family depending algebraically on a parameter t .

Although G is enormous, its existence makes it easier to handle “motivic” objects.

Example 1. The category $\text{Ind } \mathcal{M}(k)$ of ind-objects of the category $\mathcal{M}(k)$ of motives over k inherits the tensor product of $\mathcal{M}(k)$. This allows one to define a commutative Hopf algebra in $\text{Ind } \mathcal{M}(k)$ as an object H endowed with a product and coproduct, $H \otimes H \rightarrow H$ and $H \rightarrow H \otimes H$, and also with unit, counit, and antipode, $\mathbf{1} \rightarrow H$, $H \rightarrow \mathbf{1}$, $H \rightarrow H$, satisfying the appropriate axioms. The category of motivic affine group schemes is defined to be the opposite of the category of commutative Hopf algebras in $\text{Ind } \mathcal{M}(k)$. If ω is a fiber functor with values in $\mathbb{Q}$ -vector spaces V , with motivic Galois group $G := \underline{\text{Aut}}^\otimes(\omega)$, then ω induces an equivalence of $\text{Ind } \mathcal{M}(k)$ with the category of linear representations—not necessarily finite-dimensional—of G , and $H \mapsto \text{Spec}(\omega(H))$ is an equivalence from the category of motivic affine group schemes to the category of affine group schemes over $\mathbb{Q}$ endowed with an action of G .

Here are three examples of such objects.

1.3.1. The motivic version of the unipotent completion of $\pi_1(X, x)$, cf. Deligne [14], especially §§5, 7, and 10–13.

1.3.2. The motivic Galois group: it has a motivic version G_{mot} such that, for every fiber functor ω , the corresponding motivic Galois group is obtained from G_{mot} by applying ω . This is a construction valid in every Tannakian category, cf. Deligne [15, §8].

In the analogy with the theory of the profinite π_1 , its analogue is the following: as x varies, the $\pi_1(X, x)$ form a local system on X , or more precisely a group object in the category of pro-objects of the category of étale covers of X . The π_1 at x is obtained by taking the fiber at x .

1.3.3. Let $\mathcal{A}$ be a full subcategory of the category of mixed motives over k , stable under $\otimes$, duals, and subquotients, and let $\mathcal{A}_{\text{ss}}$ be the full subcategory of direct sums of pure motives in $\mathcal{A}$. There is an exact tensor-compatible functor $M \mapsto \text{Gr}^W(M)$ from $\mathcal{A}$ to $\mathcal{A}_{\text{ss}}$. There exists in $\text{Ind } \mathcal{A}_{\text{ss}}$ a motivic pro-unipotent group scheme U , acting functorially on $\text{Gr}^W(M)$ for M in $\mathcal{A}$, such that $M \mapsto \text{Gr}^W(M)$ is an equivalence of $\mathcal{A}$ with the category of representations, in $\mathcal{A}_{\text{ss}}$, of U . This is a consequence of Deligne [15, 8.17].

One may define and handle motivic schemes, torsors, and so on in the same way.

1.3 The abelianized motivic Galois group

The abelianization G^{ab} of the motivic Galois group has a more reasonable size. It is independent of the chosen realization functor, and even of the existence of such a functor. Conjecturally, if $k_1 \subset k_2$ are two algebraically closed fields, G^{ab} is the same for k_1 and k_2 , and the corresponding subcategory of motives is generated by the H_{mot}^1 of abelian varieties of CM type.

In characteristic 0, if motives are taken in the sense of absolute Hodge cycles and one restricts to the subcategory generated by the H^1 of abelian varieties, one can compute G^{ab} . For $k = \overline{\mathbb{Q}}$, one can even compute the quotient of G by the derived group of $\text{Ker}(G \rightarrow \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}))$. See Deligne [13] or Schappacher's lecture at this conference.

For $k = \overline{\mathbb{F}}_p$, every motive should be a direct sum of pure motives and the motivic Galois group G should contain an element F , Frobenius, whose powers are dense in G . In particular, G should be commutative. The conjectural calculation of $\mathcal{M}(\overline{\mathbb{F}}_p)$ was made by Grothendieck (unpublished). See Langlands–Rapoport [23] and Milne's lecture at this conference.

1.4 Abelian varieties and 1-motives

One source for 1.1(a), (b) is the following example. A smooth projective curve X defines its Jacobian $J(X) = \text{Pic}^0(X)$. Abelian varieties up to isogeny over k form a semisimple abelian category with $\text{Hom}(A, B)$ finite-dimensional over $\mathbb{Q}$, and for each usual cohomology theory H , the dual $H_1(X)$ of $H^1(X)$ is obtained from $J(X)$ by applying a “realization” functor. Note that $H_1(X)$ is also the Tate twist $H^1(X)(1)$ of $H^1(X)$. For ℓ -adic cohomology, $\text{real}_{\ell}(A)$ is the Tate module $T_{\ell}(A) = \varprojlim A_{\ell^n}(\bar{k}) \otimes \mathbb{Q}_{\ell}$. For k of characteristic 0 and de Rham theory, $\text{real}_{\text{DR}}(A)$ is the Lie algebra of the universal additive extension of A .

Thus one wants to be able to identify abelian varieties with particular motives of weight -1 . For some applications, such as the study of Shimura varieties, it would already be satisfactory to have a category of motives containing the abelian varieties and stable under $\otimes$. In characteristic 0, the theory of absolute Hodge cycles provides such a theory (Deligne–Brylinski [11], Deligne–Milne [12], or Panchishkin's lecture at this conference). If one does not restrict to pure motives, another key example is that of smooth curves which are not necessarily complete. More generally, one may consider a connected smooth projective curve Y , disjoint subsets S and T of Y , and the cohomology $H^1(Y - S, \text{rel}T)$. A 1-motive K is a complex of group schemes concentrated in degrees -1 and 0 , with, over the algebraic closure, K^{-1} a free finitely generated $\mathbb{Z}$ -module and K^0 an extension of an abelian variety by a torus. Assume for simplicity that k is algebraically closed. For each usual cohomology theory, the H^1 under consideration, or rather its Tate twist $H^1(Y - S, \text{rel}T)(1)$, is then obtained from the following 1-motive by applying a realization functor.

Let $J_{S,T}(Y)$ be the generalized Jacobian classifying degree-zero invertible sheaves on Y trivialized on T . It is an extension of the abelian variety $\text{Pic}^0(Y)$ by the torus whose character group is $\text{Ker}(\mathbb{Z}^T \rightarrow \mathbb{Z})$. Each $s \in S$ defines the invertible sheaf $\mathcal{O}(s)$, trivialized on T and of degree 1, whence

$$\text{Ker}(\mathbb{Z}^S \rightarrow \mathbb{Z}) \rightarrow J_{S,T}(Y). \quad (1.5.1)$$

This is the promised 1-motive. See Deligne [9, §10].

Warning 2. For $S, T \neq \emptyset$, the 1-motive (1.5.1) also determines the category of invertible sheaves on $Y - S$, trivialized on T : a point $x \in J_{S,T}(Y)$ defines such an invertible sheaf $\mathcal{L}_x$, and an isomorphism from $\mathcal{L}_x$ to $\mathcal{L}_y$ which is the identity on T is identified with $k \in \text{Ker}(\mathbb{Z}^S \rightarrow \mathbb{Z})$, with $y - x = \delta k$.

Again, one wants to be able to identify 1-motives with particular mixed motives. For some applications, such as the study at infinity of Shimura varieties, it would already be satisfactory

to have a category of motives containing the 1-motives and stable under $\otimes$. In characteristic 0, this is known (Brylinski [7]).

1.5 Moduli of motives and Shimura varieties

Abelian varieties have moduli spaces, and these moduli spaces make it possible to interpret algebraically certain quotients of Hermitian symmetric domains by arithmetic groups. The case of motives is more complicated. If S is a scheme over $\mathbb{C}$ and M is a family of pure motives parameterized by S , then the Hodge realization of M gives a variation of Hodge structures $\mathcal{M}_{\mathbb{C}}$ on $S(\mathbb{C})$, polarized if M is polarized. Hence there is a map φ_0 from $S(\mathbb{C})$ to the space $\mathcal{H}$ classifying polarized Hodge structures with the same Hodge numbers as M .

The variation $\mathcal{M}_{\mathbb{C}}$ gives in particular a local system of complex vector spaces $\mathcal{M}_{\mathbb{C}}$ endowed with a Hodge filtration F varying continuously. The holomorphy of F and the transversality condition discovered by Griffiths [17] (see also Griffiths [18]) may be expressed as follows. Let T be the tangent bundle of S , viewed as a C^∞ variety. The complex structure of S gives a complex structure on T . Let $\varphi : T \otimes_{\mathbb{R}} \mathbb{C} \rightarrow T$ be the $\mathbb{C}$ -linear extension of the identity of T , and let $T'' \subset T \otimes_{\mathbb{R}} \mathbb{C}$ be the kernel of φ . Define the Hodge filtration of $T \otimes \mathbb{C}$ by

$$F^{-1} = T \otimes \mathbb{C}, \quad F^0 = T'', \quad F^1 = 0.$$

For ξ and η smooth sections of $T \otimes \mathbb{C}$ and of $\mathcal{M}_{\mathbb{C}}$, the flat structure of $\mathcal{M}_{\mathbb{C}}$ allows one to define $\nabla_{\xi}\eta$. The condition is that $\xi, \eta \mapsto \nabla_{\xi}\eta$ be compatible with the Hodge filtrations:

ξ in T'' , η in F^i , $\nabla_{\xi}\eta$ in F^i : holomorphy;

ξ in T , η in F^i , $\nabla_{\xi}\eta$ in F^{i-1} : transversality.

On the classifying space $\mathcal{H}$ there is a complex structure and a holomorphic distribution $\tau \subset T$ such that holomorphy and transversality translate into holomorphy of the classifying map $\varphi : S(\mathbb{C}) \rightarrow \mathcal{H}$ and tangency of $\tau(S)$ to τ .

The distribution τ is generally nonintegrable. The set of points of $\mathcal{H}$ corresponding to a direct factor of some $H^i(X)$, with X nonsingular projective, is a countable union of subvarieties tangent to τ . If τ is not the entire tangent bundle, we have no description of this set, even conjectural.

An analogous argument, also due to Griffiths, explains why the points of the intermediate Jacobian $J^d(X)$ cannot all be classes of algebraic cycles, except when $H^{2d-1}(X)$ is of type $\{(d-1, d), (d, d-1)\}$. Let $f : X \rightarrow S$ be a family of nonsingular projective varieties, and put $X_s = f^{-1}(s)$. The intermediate Jacobians $J^d(X_s) = H^{2d-1}(X_s, \mathbb{C})/F^d + H^{2d-1}(X_s, \mathbb{Z})$ form a holomorphic bundle J^d of complex tori, and the transversality axiom ensures that the Gauss–Manin connection, expressing the fact that the $H^{2d-1}(X_s, \mathbb{C})$ form a local system, descends to the quotient to define a differential operator D on the sheaf of sections of J^d , with values in the sheaf of 1-forms valued in the holomorphic bundle $H^{2d-1}(X_s, \mathcal{O})/F^{d+1}$. An algebraic family $(Z_s)_{s \in S}$ of algebraic cycles homologous to zero on the X_s defines a holomorphic section z of J^d . Griffiths’s result is that $Dz = 0$. The section z defines a holomorphic family of mixed Hodge structures, an extension of $\mathbb{Z}$ of type $(0, 0)$ by $H^{2d-1}(X)(d)$, and $Dz = 0$ is equivalent to this family satisfying the transversality axiom.

1.6

When the distribution τ is the entire tangent bundle, $\mathcal{H}$ is an arithmetic quotient of a Hermitian symmetric domain. This description of the complex points of Shimura varieties as moduli of Hodge structures suggests that these varieties are moduli spaces of motives.

Let $\mathbb{S}$ be the real algebraic group $\mathbb{C}^*$: $\mathbb{S} = R_{\mathbb{C}/\mathbb{R}}\mathbb{G}_m$. Giving an action of $\mathbb{S}$ on the real vector space V amounts to giving a decomposition of $V \otimes \mathbb{C}$ as a direct sum of $V^{p,q}$ with $V^{q,p} = \overline{V^{p,q}}$,

where $z \in \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ acts on $V^{p,q}$ by multiplication by $z^{-p}\bar{z}^{-q}$. Define $w : \mathbb{G}_m \rightarrow \mathbb{S}$, respectively $\mu : \mathbb{G}_m \rightarrow \mathbb{S}_{\mathbb{C}}$, defined over $\mathbb{C}$, by the condition that, for every V , $w(\lambda)$, respectively $\mu(\lambda)$, acts on $V^{p,q}$ by multiplication by λ^{p+q} , respectively λ^{-p} .

The data defining a Shimura variety $\mathrm{Sh}_K(G, X)$ are a reductive group G over $\mathbb{Q}$, a $G(\mathbb{R})$ -conjugacy class of morphisms $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$, with $w_h := h \circ w$ central (hence independent of h), and an open compact subgroup K of $G(\mathbb{A}^f)$. Put ourselves in the case where w_X is defined over $\mathbb{Q}$ and $\mathrm{Int} h(i)$ is a Cartan involution of the quotient $G/w_X(\mathbb{G}_m)$.

The reflex field $E(G, X)$ is the field of definition of the conjugacy class of μ_h , for arbitrary $h \in X$. It is a subfield of $\mathbb{C}$. The Shimura variety is defined over $E(G, X)$, with complex points $K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q})$. Its construction in the general case is due to Borovoi [6].

A point of the Shimura variety over a field F containing $E(G, X)$ should correspond to

- (a) an exact tensor functor χ from the category $\mathrm{Rep}(G)$ of representations of G to the category of pure motives over F ;
- (b) an integral structure. In terms of the finite adelic realization, the restricted product of the ℓ -adic realizations, it can be described as an isomorphism of tensor functors

$$\chi(V)_{\mathbb{A}^f} \xrightarrow{\sim} V \otimes \mathbb{A}^f,$$

given up to composition with an element of K .

The following condition must be satisfied. For simplicity, assume here that F can be embedded in $\mathbb{C}$.

- (c) Let σ be an embedding of F into $\mathbb{C}$, extending the identity embedding of $E(G, X)$ into $\mathbb{C}$. There must exist $h \in X$ such that the following tensor functors from $\mathrm{Rep}(G)$ to Hodge structures are isomorphic:
 - (1) $(V, \rho) \mapsto V$ endowed with the Hodge structure defined by $\rho \circ h$;
 - (2) $(V, \rho) \mapsto$ the Hodge realization of $\chi(V)$, after extension of the base field to $\mathbb{C}$ by σ .

That this condition (c) be independent of the chosen complex embedding σ —assumed to extend the inclusion of $E(G, X)$ into $\mathbb{C}$ —is not evident. Sometimes, however, it follows from (b) and from the more algebraic conditions (d), (e), consequences of (c), that follow.

- (d) The de Rham realization defines a fiber functor on $\mathrm{Rep}(G)$: $V \mapsto \chi(V)_{\mathrm{DR}}$, corresponding to a G -torsor P over F . The Hodge filtration of the $\chi(V)_{\mathrm{DR}}$ is exact and compatible with the tensor product, hence comes from a parabolic Q of P and from $\mu_{\mathrm{DR}} : \mathbb{G}_m \rightarrow$ the center of $Q/\mathrm{Ker}(Q \rightarrow P)$, lifting to a conjugacy class of morphisms from $\mathbb{G}_m$ to Q . Since F contains $E(G, X)$, it makes sense to ask that the corresponding conjugacy class of maps from $\mathbb{G}_m$ to G coincide with that of the $h \circ \mu$ ($h \in X$). Condition (c) implies this. This explains the appearance of the reflex field.
- (e) For (V, ρ) a representation of weight 0—that is, $\rho \circ w_h$ is trivial—and $V \otimes V \rightarrow \mathbb{Q}$ an invariant symmetric bilinear form such that, on $V_{\mathbb{R}}$, $B(v, h(i) \cdot w)$ is symmetric positive definite, one asks that $\chi(V) \otimes \chi(V) \rightarrow \mathbf{1}$ be positive for the expected polarization of the category of motives (loc. cit. V, 2.4, p. 276).

1.7

This motivic interpretation of Shimura varieties was a guide in developing the axioms that characterize them, and also in determining their conjugates (Borovoi [6]).

Let p be a prime number such that $G_{\mathbb{Q}_p}$ extends to a reductive group over $\mathbb{Z}_p$, with K the product of $G(\mathbb{Z}_p)$ with a subgroup of the restricted product of the $G(\mathbb{Q}_{\ell})$, $\ell \neq p$. The reflex field $E(G, X)$ is then unramified at p .

One expects the Shimura variety to have a natural reduction modulo p , and one may try to paraphrase the preceding conjectural motivic description in order to conjecture the structure of its set of points over a finite field of characteristic p . There are difficulties: how should one interpret (c) and the p -part of (b), which must be treated jointly with (d)? See, in this direction, Langlands–Rapoport [23].

2 Cohomology theories

2.1 Transfer principle

Principle. A geometric construction possible in one of the usual cohomology theories should have a motivic meaning and therefore have an analogue in the other usual theories.

This principle was crucial in developing the theory of mixed Hodge structures. Grothendieck saw that every $H^i(X)$ should be devissaged in terms of subquotients of the cohomology of nonsingular projective varieties. The mixed motive $H_{\text{mot}}^i(X)$ should therefore have a class in the Grothendieck group of pure motives. In characteristic 0, the Hodge number $h^{p,q}$ of the Hodge realization of a pure motive defines a homomorphism from this Grothendieck group to $\mathbb{Z}$. Thus Hodge numbers $h^{p,q}(H_{\text{mot}}^i(X))$ should have a meaning.

To go further, one had to be convinced that every motive has an increasing weight filtration W with $\text{Gr}_i^W(M)$ pure of weight i —that is, a direct factor of $H_{\text{mot}}^i(X)$ for X nonsingular projective. It was from the H^1 of curves, i.e. from 1-motives, that I became convinced of this; the first test that the definition of mixed Hodge structures had to pass was that they recover as a special case the 1-motives over $\mathbb{C}$ (Deligne [9, §10]).

The same transfer principle made it possible to conjecture the asymptotic behavior of a variation of Hodge structures on a punctured disk, or on a product of punctured disks.

Warning 3. Let (S, η, s) be a trait, with S the spectrum of a complete discrete valuation ring. According to Raynaud, an abelian variety over the generic point η , with semistable reduction, admits a rigid analytic description as the cokernel of a map defining a 1-motive. In parallel, in Hodge theory, if $\mathcal{F}$ is a polarized variation of Hodge structures of type $\{(1,0), (0,1)\}$ on a punctured disk D^* , the monodromy gives on $\mathcal{F}_0$ a weight filtration W which, together with the original Hodge filtration, makes $\mathcal{F}$ a variation of mixed Hodge structures in a punctured neighborhood of 0.

One should not expect analogous behavior for motives when the transversality condition is nontrivial in Hodge theory. In general, an object describing the asymptotic behavior should exist only on the punctured Zariski tangent space (assuming semistable reduction), and it should not suffice to reconstruct the original object. For example, for $\mathcal{F}$ a polarized variation of Hodge structures on $D^* = D - \{0\}$, W. Schmid’s asymptotic nilpotent orbit is a variation of mixed Hodge structures on the punctured tangent space at 0, and, when $\mathcal{F}$ extends over D , it is the constant variation with value the fiber at 0 of the extension.

Morihiro Saito’s theory (Saito [29]), which provides the six operations and vanishing cycles in mixed Hodge theory, is partly inspired by ℓ -adic theory and the motivic viewpoint. Conversely, it suggests that if one wants to consider motives over a base S , then rather than seeking motives M whose ℓ -adic realizations are ℓ -adic sheaves on S , it may be preferable to have motives whose realizations are perverse sheaves. In any case, only in such a framework can one hope for a weight filtration.

It is again the philosophy of motives that led Grothendieck to conjecture the existence of the “mysterious” functor relating the p -adic étale cohomology of a variety defined over $\mathbb{Q}_p$, assumed to have good reduction, to its de Rham cohomology: since it existed for H^1 , i.e. for the motives that are abelian varieties, it had to exist always, compatibly with the tensor product.

2.2 The motivic π_1

Let X_0 be an algebraic variety over k and let X be obtained from X_0 by extension of scalars to an algebraic closure $\bar{k}$ of k . For $x \in X_0(k)$, the profinite fundamental group $\pi_1(X, x)$ is endowed with an action of $\text{Gal}(\bar{k}/k)$. Assume for simplicity that X is normal. The functor “fiber at x ” is then an equivalence of categories from lisse $\mathbb{Q}_\ell$ -sheaves on X_0 to ℓ -adic representations of $\text{Gal}(\bar{k}/k)$ endowed with an equivariant action of $\pi_1(X, x)$.

I know of no sense in which $\pi_1(X, x)$ is motivic. The situation changes for its pro- ℓ completion $\pi_1(X, x)^{(\ell)}$. Its abelianization is $H_1(X, \mathbb{Z}_\ell)$, and its H_2 is contained in that of X . There should exist a motivic group scheme $\pi_1(X, x)^{\text{mot}}$ over k such that

- (a) for each usual cohomology theory H , the corresponding realization of $\pi_1(X, x)^{\text{mot}}$ over $\mathbb{Q}$ has as its linear representations the “local systems” on X , in the sense of H , which are unipotent;
- (b) the functor “fiber at x ” is an equivalence of categories:

$$\begin{array}{c} \text{smooth motives over } X_0, \text{ iterated extensions of inverse images of motives over } k \\ \xrightarrow{\sim} \\ \text{motives over } k \text{ with an action of } \pi_1(X, x)^{\text{mot}}. \end{array}$$

In (a), the local systems considered are not assumed to be motivic. According to (b), those that are motivic nevertheless form a faithful system of representations.

This philosophy inspired the Hodge theory of π_1 : see Steenbrink–Zucker [30].

More generally, suppose that a set A of pure motives over X_0 is given. The preceding case corresponds to $A = \mathbb{Z}$, or, what is the same, to A reduced to the unit motive. There should exist over k a motivic group scheme $\pi_1(X, x)_A$, the motivic π_1 relative to A , such that

- (a') over $\bar{k}$, its relative realization for the theory H has as its representations the “local systems” on X which are iterated extensions of subquotients of tensor products of realizations of objects of A and of their duals;
- (b') the functor “fiber at x ” is an equivalence from smooth motives over X , which are iterated extensions of motivic subquotients of tensor products of objects of A , of their duals, and of inverse images of motives over k , to motives over k endowed with an action of $\pi_1(X, x)_A$.

In (b'), the subquotients are to be taken over X and are not assumed to be motivic.

In Hodge theory, this suggests the following question. Let X be smooth and connected over $\mathbb{C}$, and let $x \in X$. Let A be a set of polarizable variations of Hodge structure on X . Let $\mathcal{V}_A$ be the Tannakian category of $\mathbb{Q}$ -local systems on X generated, under $\otimes$, duals, subquotients, and extensions, by the local systems underlying the objects of A . Let $\mathcal{M}_A$ be the Tannakian category of admissible mixed variations of Hodge structure on X , in the sense of Steenbrink–Zucker [30], generated, under $\otimes$, duals, subquotients, and extensions, by A and by the constant variations whose associated graded for the weight filtration is polarizable.

On $\mathcal{M}_A$ one has the fiber functor “fiber at x ”, denoted ω_x . Let $G = \text{Aut}^\otimes(\omega_x)$. It is the Zariski closure of $\pi_1(X, x)$ in the product of the $\text{GL}(H_x)$, for H an object of $\mathcal{M}_A$. Let $\mathcal{O}(G)$ be the Hopf algebra whose spectrum is G .

For M in $\mathcal{M}_A$, the underlying local system $M_{\mathbb{Q}}$ belongs to $\mathcal{V}_A$, so that G acts on its fiber $(M_{\mathbb{Q}})_x$ at x : this fiber is an $\mathcal{O}(G)$ -comodule,

$$(M_{\mathbb{Q}})_x \rightarrow (M_{\mathbb{Q}})_x \otimes \mathcal{O}(G). \quad (2.2.1)$$

The Hopf algebra $\mathcal{O}(G)$ should carry an ind-mixed Hodge structure. The group G should be an affine group scheme in the Tannakian category of mixed Hodge structures, so that: (a)

the maps (2.2.1) are morphisms of mixed Hodge structures; (b) the functor $M \mapsto M_x$ is an equivalence of $\mathcal{M}_A$ with the category of mixed Hodge structures H whose associated graded is polarizable, endowed with a comodule structure $H \rightarrow H \otimes \mathcal{O}(G)$ which is a morphism of mixed Hodge structures.

3 Absolute cohomology

3.1

In each of the usual cohomology theories, the functors $H^i(X)$ arise from a functor $R\Gamma(X)$ with values in a triangulated category $\mathcal{D}$. This category is endowed with a t -structure, namely the datum of subcategories $\mathcal{D}^{\leq 0}$ and $\mathcal{D}^{\geq 0}$ satisfying appropriate axioms (Beilinson–Bernstein–Deligne [3, 1.3]), whose heart, the abelian subcategory $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$, is the category in which H^0 takes its values, and

$$H^i(X) = H^i R\Gamma(X)$$

in the sense of this t -structure.

One may hope that these functors $R\Gamma$ are the “realizations” of a functor $R\Gamma_{\text{mot}}$ with values in a triangulated category $\mathcal{DM}(k)$, with t -structure whose heart is the category $\mathcal{M}(k)$ of mixed motives over k , such that the $R\Gamma$ in the various theories are obtained by applying realization functors.

A triangulated category $\mathcal{D}$ with a t -structure is not always the derived category of its heart $\mathcal{E}$. Example: let X be a finite connected CW-complex and let $\mathcal{D} \subset D^b(X, \mathbb{Q})$ be the subcategory of the derived category formed by complexes whose cohomology sheaves are locally constant. Its heart is the abelian category of locally constant sheaves of $\mathbb{Q}$ -vector spaces on X . There is a natural functor $D^b(\mathcal{E}) \rightarrow \mathcal{D}$, but this functor is an equivalence only if X is a $K(\pi, 1)$.

In this section, we shall assume the existence of $\mathcal{DM}(k)$ and of “realization” functors. We shall also assume the existence in $\mathcal{DM}(k)$ of a “Tate twist” which is an auto-equivalence of categories and which corresponds, under the realization functors, to the usual Tate twist. Finally, we shall assume that the category $\mathcal{M}(k)$ is Tannakian and that ℓ -adic cohomology is a fiber functor. This implies that if $\varphi : M_1 \rightarrow M_2$ is a morphism of motives whose ℓ -adic realization is an isomorphism, then φ is an isomorphism.

For M_1, M_2 in $\mathcal{M}(k)$, put

$$\text{Ext}^i(M_1, M_2) = \text{Hom}_{\mathcal{DM}(k)}(M_1, M_2[i]).$$

These are the usual Hom and Ext for $i = 0, 1$, but for $i > 1$ they may differ from Yoneda Ext groups, which are the $\text{Hom}(M_1, M_2[i])$ in $D^b(\mathcal{M}(k))$.

For each of the usual cohomology theories, these motivic Ext groups should map to the Ext^i groups between the corresponding realizations of M_1 and M_2 .

3.2

For an algebraic variety X over k , call the following the **absolute cohomology groups** of X :

$$H_{\text{abs}}^i(X, \mathbb{Q}(j)) := \text{Hom}_{\mathcal{DM}(k)}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[i]),$$

where $\mathbf{1}$ is the unit motive. These groups are often called **motivic cohomology groups**. They are vector spaces over $\mathbb{Q}$. One should take care not to confuse them with the $H_{\text{mot}}^i(X)$, which are motives over k . The two are related by a spectral sequence

$$E_2^{p,q} = \text{Ext}^p(\mathbf{1}, H_{\text{mot}}^q(X)(j)) \Rightarrow H_{\text{abs}}^{p+q}(X, \mathbb{Q}(j)). \quad (3.2.1)$$

For a motive M , we shall also write $H_{\text{abs}}^i(M)$ for $\text{Ext}^i(\mathbf{1}, M) = \text{Hom}_{\mathcal{DM}(k)}(\mathbf{1}, M[i])$. With this notation, the spectral sequence (3.2.1) can be rewritten as

$$E_2^{p,q} = H_{\text{abs}}^p(H_{\text{mot}}^q(X)(j)) \Rightarrow H_{\text{abs}}^{p+q}(X, \mathbb{Q}(j)). \quad (3.2.2)$$

The ℓ -adic analogue is the following. A variety X over k , with algebraic closure $\bar{k}$, defines ℓ -adic cohomology groups $H^i(X \otimes_k \bar{k}, \mathbb{Q}_\ell)$, which one hopes are realizations of motives $H_{\text{mot}}^i(X)$. The variety X also has absolute etale cohomology $H^i(X, \mathbb{Q}_\ell)$, which is the abutment of a spectral sequence

$$H^p(\text{Gal}(\bar{k}/k), H^q(X \otimes_k \bar{k}, \mathbb{Q}_\ell)) \Rightarrow H^{p+q}(X, \mathbb{Q}_\ell).$$

For X smooth and projective over k , one expects $R\Gamma_{\text{mot}}(X)$ to be noncanonically isomorphic, in $\mathcal{DM}(k)$, to the direct sum of the $H_{\text{mot}}^i(X)[-i]$. This would follow from the following hypothesis:

For $\mathcal{L}$ an invertible sheaf on X , one has $c_1(\mathcal{L}) : R\Gamma_{\text{mot}}(X) \rightarrow R\Gamma_{\text{mot}}(X)(1)[2]$, whose ℓ -adic realization gives cup product with the Chern class $c_1(\mathcal{L})$ of $\mathcal{L}$: $H^i(X \otimes_k \bar{k}, \mathbb{Q}_\ell) \rightarrow H^{i+2}(X \otimes_k \bar{k}, \mathbb{Q}_\ell(1))$.

Indeed, take $\mathcal{L}$ ample. Since the ℓ -adic realization functor is a fiber functor, it follows from (*) and from the hard Lefschetz theorem that, for X purely of dimension N , the maps

$$c_1(\mathcal{L})^i : H_{\text{mot}}^{N-i}(X) \rightarrow H_{\text{mot}}^{N+i}(X)(i)$$

are isomorphisms. One can then apply Deligne [8].

Analyzing the proof, one can deduce, for each ample invertible sheaf $\mathcal{L}$, a canonical decomposition, indeed several canonical decompositions,

$$\eta_{\mathcal{L}} : R\Gamma_{\text{mot}}(X) \xrightarrow{\sim} \bigoplus H_{\text{mot}}^i(X)[-i].$$

See Deligne, “Decompositions in the derived category,” in this volume.

Warning 4. Since these arguments depend on the hard Lefschetz theorem, they are essentially rational. If one had an integral variant of $\mathcal{DM}(k)$, one should expect that, in that framework, $R\Gamma(X)$ is not always the direct sum of the $H^i(X)[-i]$.

3.3 Motivic classes of cycles

Let X be smooth over k . The various “classes” of an algebraic cycle Z of codimension d should come from a motivic class

$$cl(Z) \in \text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(d)[2d]) = H_{\text{abs}}^{2d}(X, \mathbb{Q}(d)). \quad (3.3.1)$$

The class (3.3.1) should depend only on the linear-equivalence class of Z .

If X is smooth and projective and one has chosen a decomposition

$$R\Gamma_{\text{mot}}(X) \xrightarrow{\sim} \bigoplus H_{\text{mot}}^i(X)[-i] \quad (*)$$

(cf. 3.2), the class (3.3.1) is equivalent to a sequence of classes

$$cl_i(Z) \in \text{Ext}^{2d-i}(\mathbf{1}, H_{\text{mot}}^{2d-i}(X)(d)) = H_{\text{abs}}^{2d-i}(H_{\text{mot}}^{2d-i}(X)(d)). \quad (3.3.2)$$

The first of these classes that is nonzero is independent of the chosen decomposition.

Example 5. In Hodge theory, Z first defines an integral cohomology class of type (d, d) , that is, a morphism of Hodge structures

$$\mathbb{Z} \rightarrow H^{2d}(X)(d).$$

If this class is zero, Z has a class in the intermediate Jacobian

$$J^d(X) = H^{2d-1}(X, \mathbb{C})/F^d + H^{2d-1}(X, \mathbb{Z}),$$

a group which is precisely $\text{Ext}^1(\mathbb{Z}, H^{2d-1}(X)(d))$ in the category of mixed Hodge structures. The story stops there, because the abelian category of rational mixed Hodge structures has cohomological dimension 1.

Example 6. Let X be smooth and projective, purely of dimension N , over an algebraically closed field k , and let $\mathcal{O}(1)$ be an ample invertible sheaf. We show how one can attach to a divisor D a class $cl(D)$ in $\text{Pic}^0(X) \otimes \mathbb{Q}$, a group that the theory of motives interprets as $H_{\text{abs}}^2(X)(1)$.

By a theorem of A. Weil [31, Theorem 7], the map from linear-equivalence classes of divisors to linear-equivalence classes of 0-cycles,

$$E \mapsto E \cdot c_1(\mathcal{O}(1))^{N-1}$$

induces an isogeny

$$\text{Pic}^0(X) \rightarrow J(X),$$

where $J(X)$ is the Albanese variety, or Jacobian, of X . We therefore have a bijection between $\text{Pic}^0(X) \otimes \mathbb{Q}$ and $J(X) \otimes \mathbb{Q}$. We take

$$cl_1(D) = \frac{\deg(c_1(D) \cdot c_1(\mathcal{O}(1))^{N-1})}{\deg(c_1(\mathcal{O}(1))^N)} \cdot c_1(\mathcal{O}(1))^N - c_1(D) \cdot c_1(\mathcal{O}(1))^{N-1}.$$

Example 7. Now work not over the spectrum of a field but over a base S , smooth over an algebraically closed field k . Let $f : X \rightarrow S$ be projective and smooth over S , and let Z be a codimension d cycle. Work in ℓ -adic cohomology, with $\mathbb{Z}$ invertible on S . The cycle Z has a class in

$$H^{2d}(X, \mathbb{Z}_\ell(d)) = \text{Hom}_{D(S)}(\mathbb{Z}_\ell, Rf_*\mathbb{Z}_\ell[2d](d)).$$

If one chooses a decomposition

$$Rf_*\mathbb{Q}_\ell \cong \bigoplus R^i f_*\mathbb{Q}_\ell[-i], \quad (3.6.1)$$

this class gives rise, as in 3.3, to a sequence of classes

$$cl_i(Z) \in \text{Ext}^i(\mathbb{Z}_\ell, R^{2d-i} f_*\mathbb{Q}_\ell(d)) = H^i(S, R^{2d-i} f_*\mathbb{Q}_\ell(d)). \quad (3.6.2)$$

When X is of the form $X_0 \times_{\text{Spec}(k)} S$, and the decomposition (3.6.1) comes from the unique analogous decomposition of the cohomology of X_0 , these classes identify with the Kunneth components of the cohomology class of Z .

3.4

Assume that X is smooth over a field. Decompose the K -theory group $K_n(X) \otimes \mathbb{Q}$ by Adams operations:

$$K_n(X) \otimes \mathbb{Q} = \bigoplus K_n(X)^{(j)},$$

where on $K_n(X)^{(j)}$ the Adams operation ψ^k acts by multiplication by k^j . Recall that $K_0(X)^{(j)}$ is the Chow group of codimension j cycles modulo rational equivalence, tensored with $\mathbb{Q}$.

The example of the usual cohomology theories leads one to hope for Chern classes

$$ch^j : K_n(X)^{(j)} \rightarrow H_{\text{abs}}^{2j-n}(X, \mathbb{Q}(j))$$

factoring through $K_n(X)^{(j)}$.

An optimistic conjecture is that the “Chern class” morphisms are isomorphisms

$$ch^j : K_n(X)^{(j)} \xrightarrow{\sim} H_{\text{abs}}^{2j-n}(X, \mathbb{Q}(j)) := \text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[2j-n]). \quad (3.7.1)$$

This conjecture underlies the terminology “absolute motivic cohomology group” for the $K_n(X)^{(j)}$, and the notation

$$H_{\mathcal{M}}^{2j-n}(X, \mathbb{Q}(j)) := K_n(X)_{\mathbb{Q}}^{(j)}.$$

Bloch [4] proposed an interpretation of the $K_n(X)^{(j)}$ as “higher Chow groups” $\text{Ch}^j(X; n)$. His argument has a gap, an insufficient moving lemma, which he promises to repair. This interpretation makes the definition of ch^j more natural, but the interpretation in terms of K -theory remains necessary for explicit calculation in the case of number fields. Higher Chow groups have the advantage of having a natural integral structure.

3.5

According to Beilinson, conjecture (3.7.1) can be deduced from hypotheses whose essential parts are: (a) the functor $X \mapsto K_*(X)$ factors through $R\Gamma_{\text{mot}}$; (b) the category $\mathcal{DM}(k)$ is the derived category of the category of motives. The conjecture has striking consequences for both sides of (3.7.1). They vanish for trivial reasons in different regions of the (i, j) -plane:

- (a) For M a motive, $\text{Ext}^i(\mathbf{1}, M) = 0$ for $i < 0$, and $\text{Ext}^0(\mathbf{1}, M) = \text{Hom}(\mathbf{1}, M)$. Hence $\text{Hom}(\mathbf{1}, R\Gamma_{\text{mot}}(X)(j)[N]) = 0$ for $N < 0$, and also for $N = 0$ and $j \neq 0$. In K -theory this gives the Beilinson–Soule conjecture that $K_n(X)^{(j)} = 0$ for $2j < n$, and also for $2j \leq n$ if $j \neq 0$.
- (b) The Chow group $\text{Ch}^j(X)$ of algebraic cycles of codimension j is zero for $j > \dim X$, and similarly $\text{Ch}^j(X; n)$ is zero for $j > \dim(X) + n$. Moreover, $\text{Ch}^j(X; n) = 0$ for $n < 0$. Applying (3.7.1), one would deduce the following vanishing statements.

Vanishings 3.8.1.

- (i) For M a pure motive of weight w , $\text{Hom}^i(\mathbf{1}, M) = 0$ for $i > -w$.
- (ii) For M effective of weight $\leq w$ and $b > 0$, $\text{Hom}^i(\mathbf{1}, M(w+b)) = 0$ for $i > w+b$.

For (i), assume M is a direct factor of $H_{\text{mot}}^a(X)(c)$, with $c \geq 0$, X a smooth projective variety. Then $\text{Ext}^i(\mathbf{1}, M)$ is a direct factor of $\text{Ch}^c(X; 2c-i-a)$, and, if it is nonzero, one has $2c-i-a \geq 0$, i.e. $i \leq 2c-a = -w$.

For (ii), one also uses the weak Lefschetz theorem (ℓ -adic, hence motivic if ℓ -adic cohomology is a fiber functor) to suppose that $\dim(X) \leq a$. Assume $c \geq a$; one must prove the vanishing of $\text{Ext}^i(\mathbf{1}, H_{\text{mot}}^a(X)(c))$ for $i > (c-a)+a = c$. This inequality is equivalent to $c > a + (2c-i-a) \geq \dim(X) + (2c-i-a)$.

That (3.7.1) is an isomorphism would also imply weak Lefschetz theorems for Chow groups, and the conjectures on Chow groups presented at this conference by J. P. Murre.

3.6

For M the unit motive, (3.8.1)(ii) asks that

$$H_{\text{abs}}^n(\text{Spec}(k), \mathbb{Q}(j)) = 0 \quad \text{for } n > j. \quad (3.9.1)$$

It is not clear to me whether one should expect an analogue over $\mathbb{Z}$ of (3.9.1). If the $H_{\text{abs}}^n(\text{Spec}(k), \mathbb{Z}(j))$ are defined as higher Chow groups, then (3.9.1) is trivially true.

A more concrete question: suppose that one has managed to define, in an “independent of ℓ ” way, Galois cohomology classes in the $H^n(\text{Gal}(\bar{k}/k), \mathbb{Z}_{\ell}(j))$. If $n > j$, (3.9.1) asks that these classes be torsion. Should one even hope that they are zero?

Example 8. Let G be an absolutely simple and simply connected algebraic group over a field k . Over the algebraic closure $\bar{k}$ of k , the relations among the cohomology of G , of BG , and of BT , for T a maximal torus, show in each of the usual cohomology theories that $H^i(G) = 0$ for $i = 1, 2$ and that $H^3(G)(2)$ has a natural generator α . Work in ℓ -adic cohomology. Over $\bar{k}$, the absolute ℓ -adic cohomology of G is the direct sum of that of the point and a reduced cohomology $\tilde{H}$, related to that of BG , through a Hochschild–Serre exact sequence. Thus $H^i(G, \mathbb{Z}_\ell(2)) = 0$ for $i \leq 2$, and $H^3(G, \mathbb{Z}_\ell(2))$ is isomorphic to $\mathbb{Z}_\ell$, with a natural generator α . This generator is primitive, hence defines a homomorphism

$$\alpha : G \rightarrow H^3(\mathrm{Spec}(k), \mathbb{Z}_\ell(2)).$$

One can verify that, as predicted by 3.9, this map is zero: by Merkurjev–Suslin [27], the H^3 is torsion-free, and a trace argument shows that the image is torsion: G splits over a finite extension k' of k , and $G(k')$ is equal to its derived group, at least if $|k'| \geq 4$.

The same circle of ideas makes it possible to construct a canonical central extension of $G(k)$ by $H^2(\mathrm{Spec}(k), \mathbb{Z}/n(2))$, which is equal to $K_2(k)/nK_2(k)$ by Merkurjev–Suslin [27]. Recall that for G split, a canonical central extension of $G(k)$ by $K_2(k)$ was constructed by Matsumoto [25, 5.11].

3.7

For a motive M over $\mathbb{Q}$, $H_{\mathrm{abs}}^i(M)$ appears in Beilinson’s conjecture on the behavior at $s = 0$ of the L -function of M : Beilinson [2] and the lectures of Soule, Nekovar, and Scholl at this conference. I refer to those lectures for a description of the conjecture and of what is known about it.

3.8

If F is a number field, (3.7.1) predicts that in the category of motives over F one has

$$\mathrm{Ext}^1(\mathbf{1}, \mathbb{Q}(j)) = K_{2j-1}(F) \otimes \mathbb{Q}, \quad (3.12.1)$$

with vanishing of the Ext^n for $n \geq 2$. The assertion (3.12.1) imposes severe restrictions on the category of motives that are iterated extensions of Tate motives, and on the systems of realizations attached to an object of this category. See Deligne [14], where one example considered in detail is that of the Lie algebra of the unipotent completion of the fundamental group of the projective line minus three points.

Here is a program for proving consequences of (3.12.1) for such systems of realizations.

- (A) Define a Tannakian category $\mathcal{T}$, whose simple objects are the tensor powers $\mathbb{Q}(n)$, $n \in \mathbb{Z}$, of a rank-one object $\mathbb{Q}(1)$, for which (3.12.1) is true “by definition.”

The idea is that Bloch’s higher Chow groups $\mathrm{Ch}^n(\mathrm{Spec}(F), i)$ are given by a simplicial complex graded by the codimension n of cycles, and that the product is almost given by a graded simplicial commutative algebra structure. “Almost,” because the intersection of two cycles is not always defined, only “generically.” Rationally, it should be possible to replace this almost algebra by a genuine commutative bigraded differential algebra, with d of degree $(1, 0)$, and

$$A^{m,n} = 0 \quad \text{if } m < 0,$$

$A^{*,0}$ reduced to $\mathbb{Q}$ in degree 0, and

$$H^i(A^{*,n}) \quad \text{reduced to } K_{2n-i}(F) \otimes \mathbb{Q} \quad \text{in degree } i.$$

One can then define $\mathcal{T}$ as a suitable category of bigraded differential modules and morphisms up to homotopy.

This part of the program was essentially carried out by May [26]. For a cubical variant, see Bloch's lecture in this volume.

- (B) Define, for each of the usual cohomology theories, a fiber functor “realization” on the Tannakian category $\mathcal{T}$ of (A).
- (C) Show that the mixed Tate motives one is interested in, for example $\mathrm{Lie} \pi_1(\mathbf{P}^1 - \{0, 1, \infty\})$ —or rather the systems of their realizations—come from an object of $\mathcal{T}$.

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